Title: Three in One Student Services (TOSS)

Problem Statement/Rationale

Organization is often recognized as a key component in a steady education, and slowly but surely, it lays the path for future success. With the hectic schedules and overflowing academic requirements, however, many scholars find it difficult to keep track of multiple other aspects of their studies (Orr, 2025). Disorderliness often inflicts stress and worry on a student, which adds once again to the importance of order.

With this in mind, the lack of said organization can be detrimental to a scholar's academic performance. For example, turning in homework late, which goes hand in hand with losing track of time, trouble with jotting down notes and prioritizing, cramming for exams last minute, and struggling overall with projects (Orr, 2026). These effects easily highlight the need for structure and order within a scholar's academic life, and even outside of it. While disorderliness can be easily undone, it's much better to target it before it starts.

Project Overview and Relevance

Three in One Student Services (TOSS) is a program designed to assist scholars in keeping their academic performance orderly and stable. The most notable features it has to offer are as follows; a requirement tracker that sorts them by importance, either by subject, deadline or type of requirement (FA, AA, ILA); a grade calculator for scholars to keep track of their marks and standing that uses user input of their raw grades and subject units; a course brochure that students can use as reference for their year/grade level's subjects; and precise error handling that validates user input.

As the world steps into a digital age, technology is being used more than ever, including in schools and general academics. Digital programs such as TOSS make assistance easily accessible for convenience. Having to perform multiple tasks such as the aforementioned ones in a physical manner can prove to be a hassle to some, which further reinforces the relevance of TOSS for the common problem of disorderliness that is commonly found in students.

Objectives

•General objectives

Develop a command-line application that assists students in managing academic tasks, including grade computation, requirement tracking, and course reference.

Specific Objectives

• Compute the General Weighted Average (GWA) based on user-input grades and subject credits.

• Convert raw percentage grades into a standardized grading scale using conditional logic.

• Collect and store subject-related data such as names, credits, and grades for processing.

• Track academic requirements by recording subject, type, and due date.

• Organize and prioritize tasks by sorting deadlines using the quicksort algorithm.

• Display requirements in order of urgency to help students manage time effectively.

• Provide a simple course brochure that lists subjects per grade level for student reference.

• Implement input validation and error handling to ensure reliable program execution

Technical Approach

The system used Python as its main programming language. It uses a menu system that contains different features for the users to select: Grade Calculator, Requirement Tracker, and Course Brochure.

The Grade Calculator allows the user to input the number of subjects, the names of the subjects, credits, and grade to calculate the General Weighted Average using a formula. The data analysis here processes these numbers to create a result that is submitted to the user.

The Requirement Tracker includes the CRUD operations. The data from the user is saved in a file and stored in the system.

Create - the user can add requirements to the list
Read - the program shows all the requirements added
Update - the user is given the ability to edit their requirements
Delete - the user can delete requirements if needed.

The Course Brochure prints/shows all courses offered and their respective course descriptions

Methodology

1. When the program is opened, the user will choose which feature they want to use for their specific requirement (1 for Grade Calculator, 2 for Requirement Tracker, 3 for Course Brochure, and 4 for exiting the program)

2. If the user chooses 1, the program asks the user how many subjects they have. The user will then input the number of subjects they are taking. If the user inputs a letter, a decimal number, or anything that's not an integer, the program will show an invalid input and will ask the user to press "Enter" to restart the process.

3. If it's valid, for how many times the user inputs the number of subjects, the program will ask for the subject name, the number of credits in that subject, and what the student's raw grade is for each subject.

4. The program will then calculate the General Weighted Average based on the data given by the user, and will return the final grade of the student. After that, the student can go back to the original page by pressing "Enter."

5. If the user chooses 2, they will be led to the guidelines on how to use the requirement tracker feature of the program. It will prompt the user to enter the subject, requirement type, and deadline. It also reminds the user to leave the space for an empty subject to finish the program. When inputting for the requirement type, if the user enters a different value than what the program gives, the program will show an invalid input.

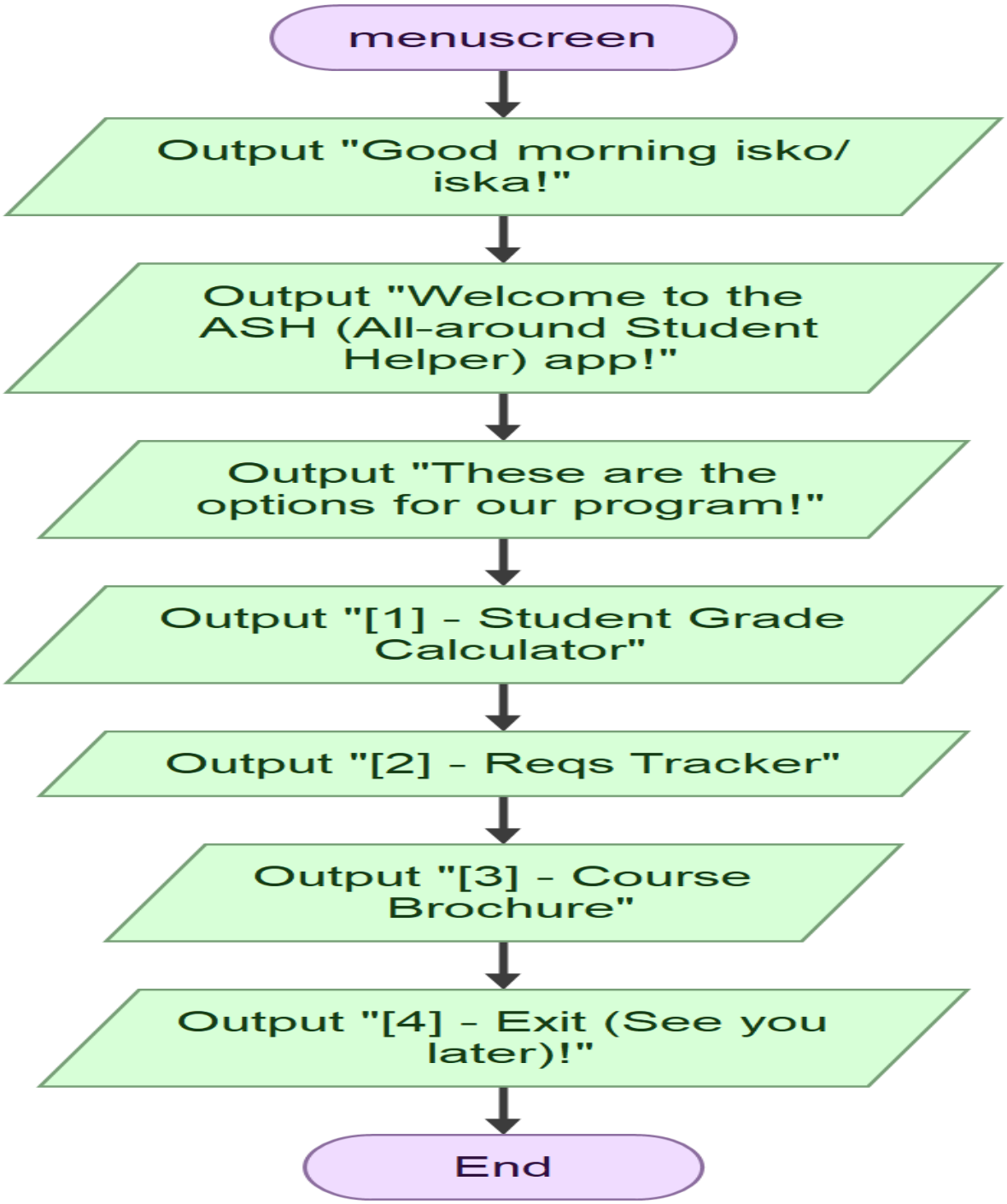
6. If input is valid, the program will ask the user to press enter to be led to the tracker itself. It will then ask for the subject, requirement type, and deadline over and over until the user leaves the subject empty. Once the user inputs an empty space, the program shows a list of all the requirements the user had previously inputted. The user can then go back to the original page by pressing "Enter".

7. If the user chooses 3, the program will output all courses offered and their respective course description. The program will also allow the user to go back to the program by pressing "Enter" once done.

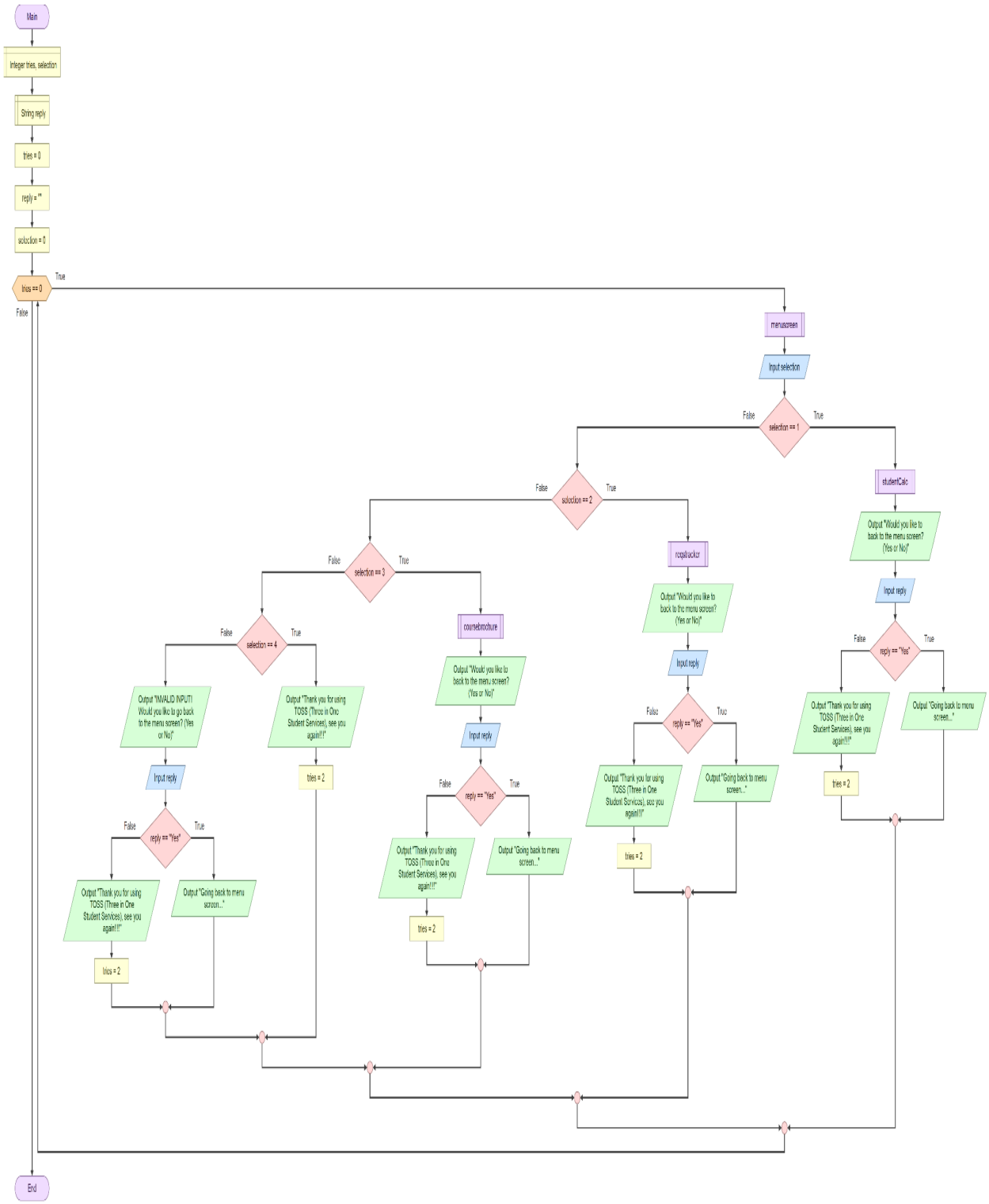
8. If the user chooses 4, the program will greet the user goodbye, and the user can press "Enter" to exit the program.

Flowchart

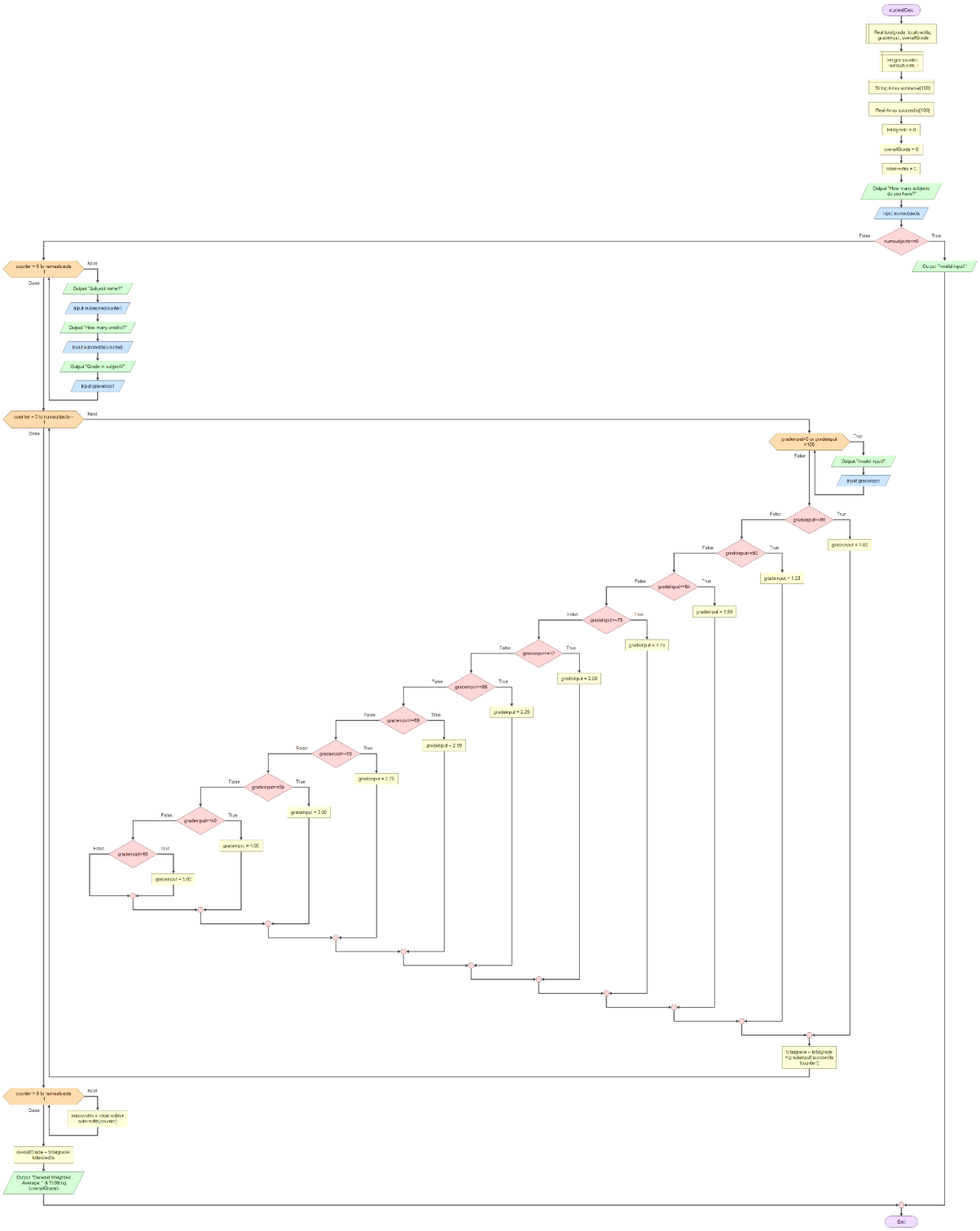
Menu Screen


menuscreen
Output "Good morning isko/ iska!"
Output "Welcome to the ASH (All-around Student Helper) app!"
Output "These are the options for our program!"
Output "[1] - Student Grade Calculator"
Output "[2] - Reqs Tracker"
Output "[3] - Course Brochure"
Output "[4] - Exit (See you later)!"
End

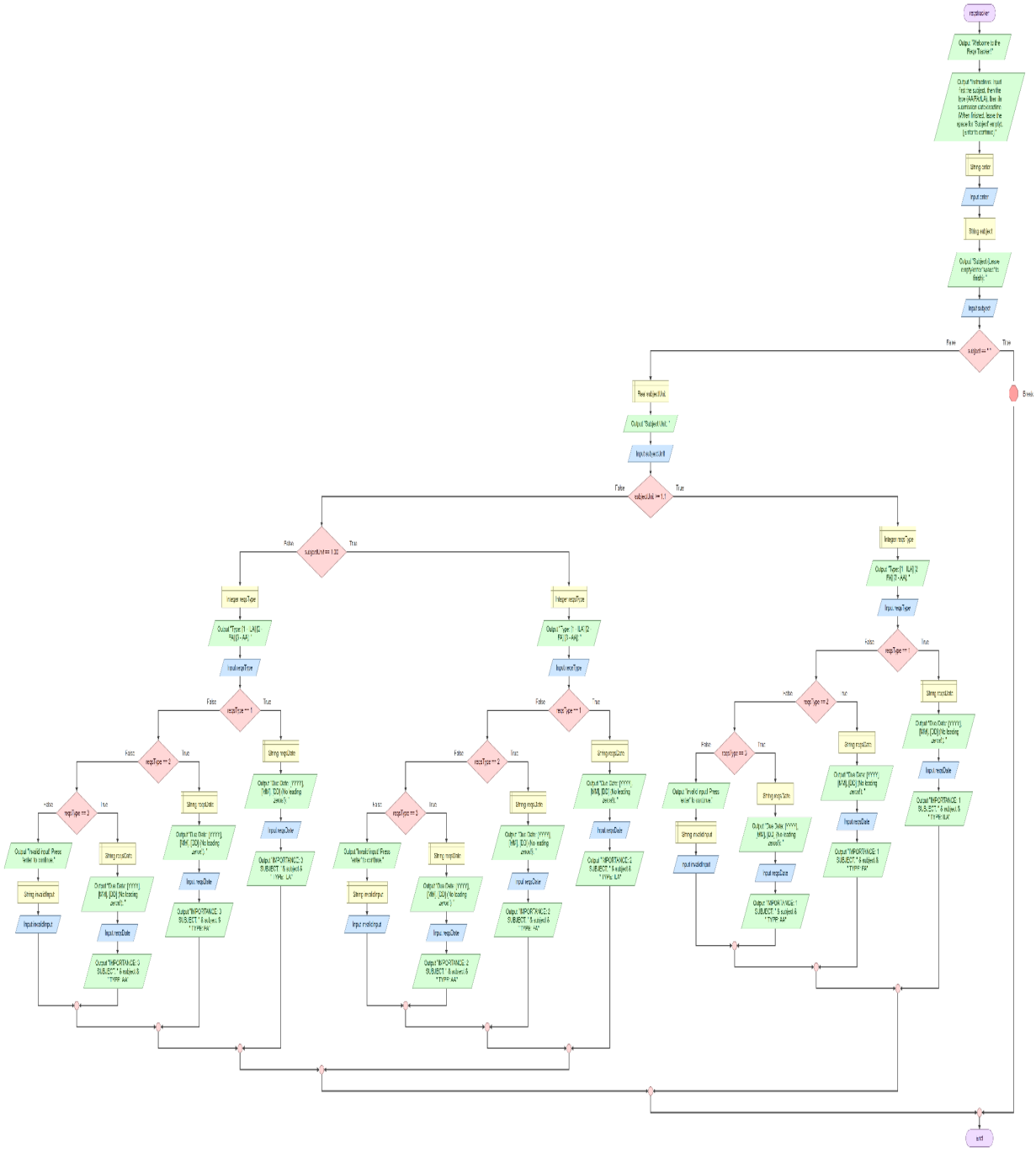
Main Loop



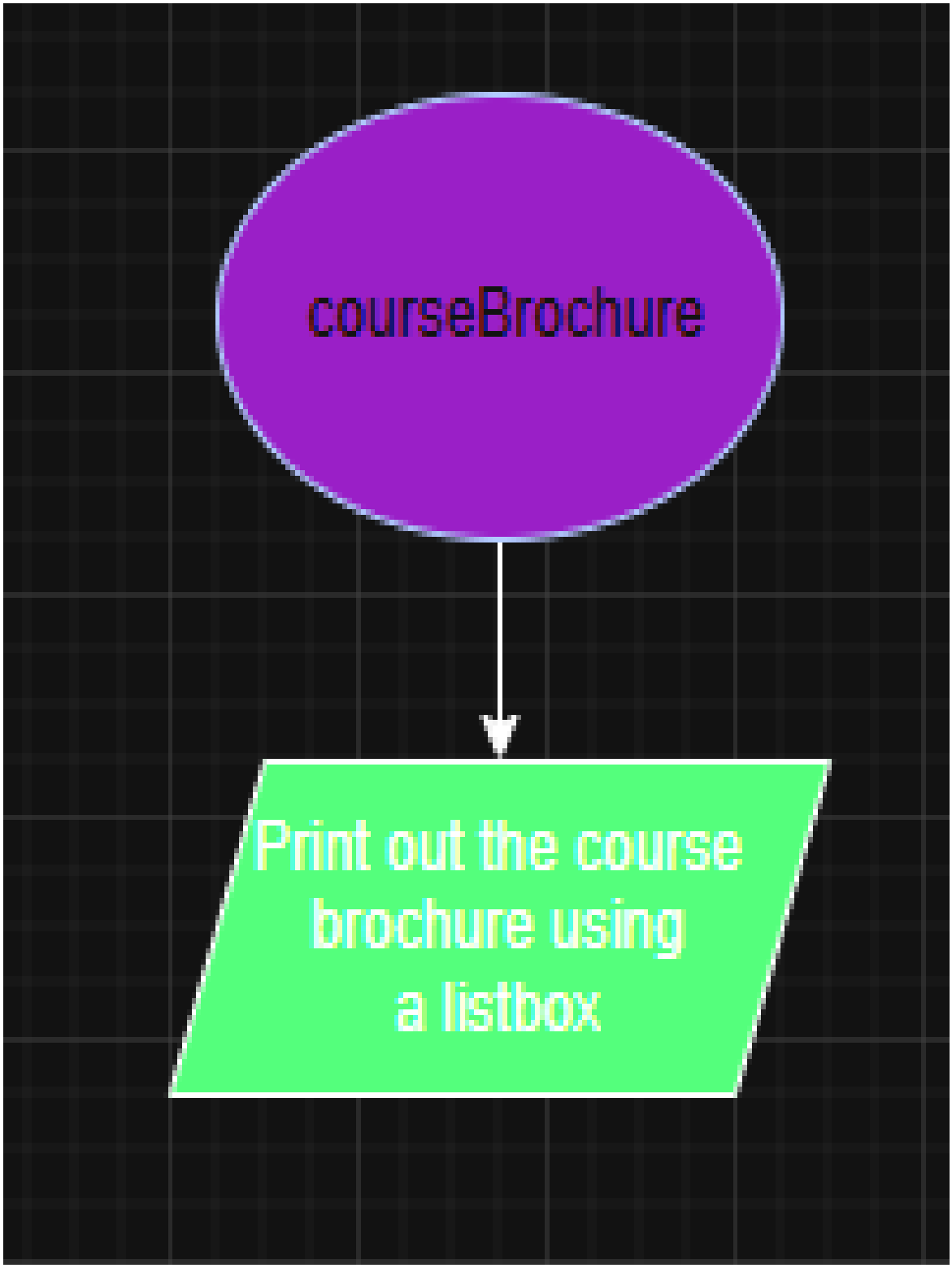
Student Calculator/GWA Calculator



Requirements Tracker

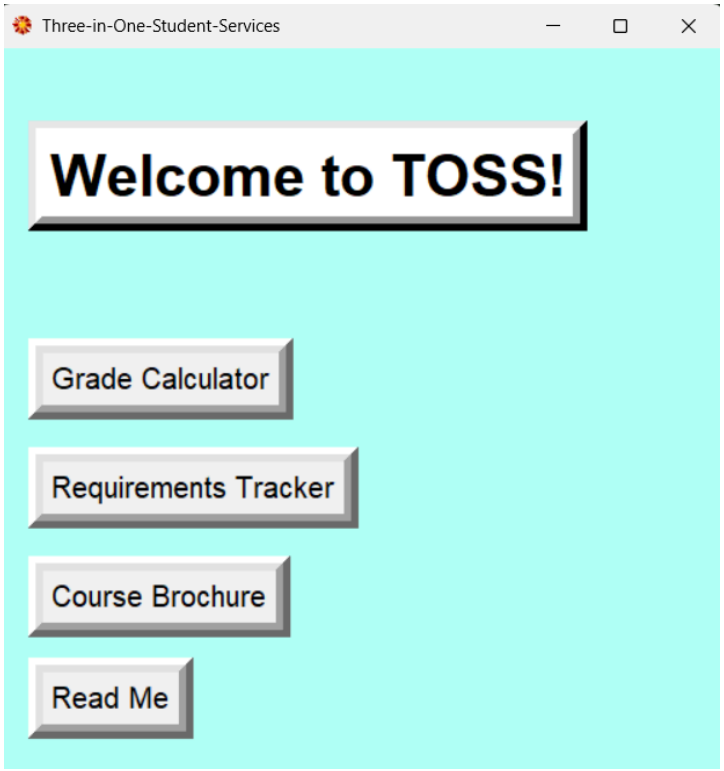

recstracker
Output "Welcome to the Recs Tracker!"
Output "Instructions: Input "recs" to see subject, the title line (AA/FA/LA), line de summaries can be written. When finished, leave the space for "Subject" empty. (press enter to continue)"
String enter
input enter
String subject
Output "Subject (Leave empty/enter "quit" to finish): "
input subject
subject == ""
False
True
Break
Free subject.Unit
Output "Subject (Unit): "
input subject.Unit
subject.Unit == 1.1
False
True
subject.Unit == 1.22
False
True
Integer recsType
Output "Type (1 - LA/12 - FA/13 - AA): "
input recsType
recsType == 1
False
True
recsType == 2
False
True
recsType == 3
False
True
String recsDate
Output "Due Date: (YYYY), (MM), (DD) (No leading zeros): "
input recsDate
Output "IMPORTANCE: 1 SUBJECT: " & subject & " TYPE: LA"
String recsDate
Output "Due Date: (YYYY), (MM), (DD) (No leading zeros): "
input recsDate
Output "IMPORTANCE: 1 SUBJECT: " & subject & " TYPE: FA"
String recsDate
Output "Due Date: (YYYY), (MM), (DD) (No leading zeros): "
input recsDate
Output "IMPORTANCE: 1 SUBJECT: " & subject & " TYPE: AA"
Output "Invalid input. Press 'enter' to continue."
String invalidInput
input invalidInput
Output "Invalid input. Press 'enter' to continue."
String invalidInput
input invalidInput
end

Course Brochure

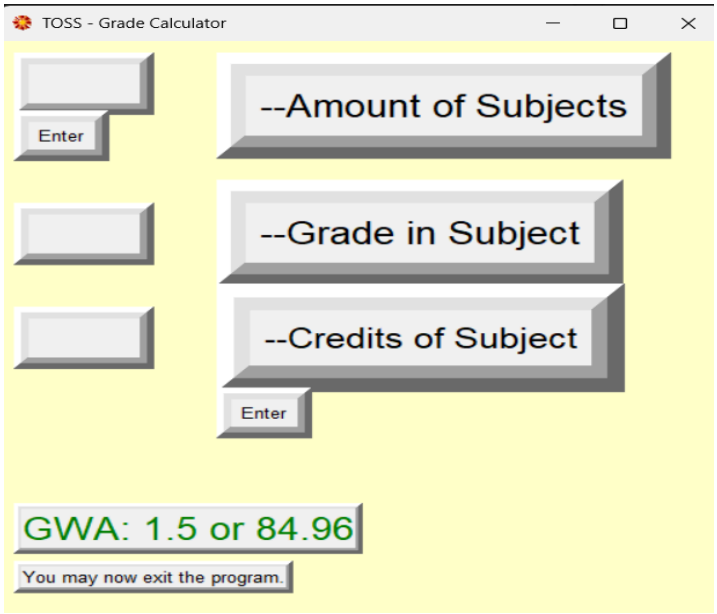


Program

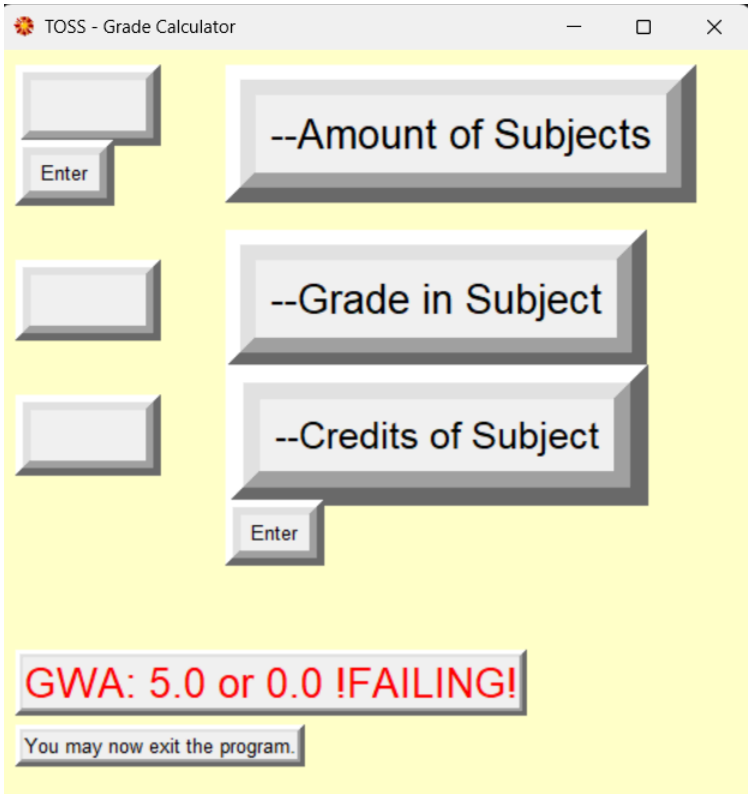
Main Menu



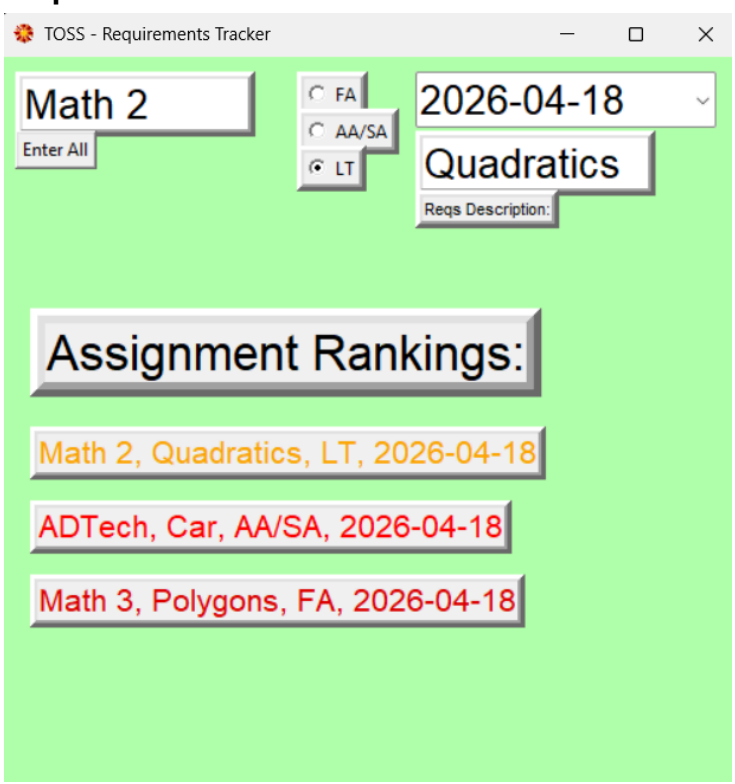
GWA Calculator (If Proper Inputs)



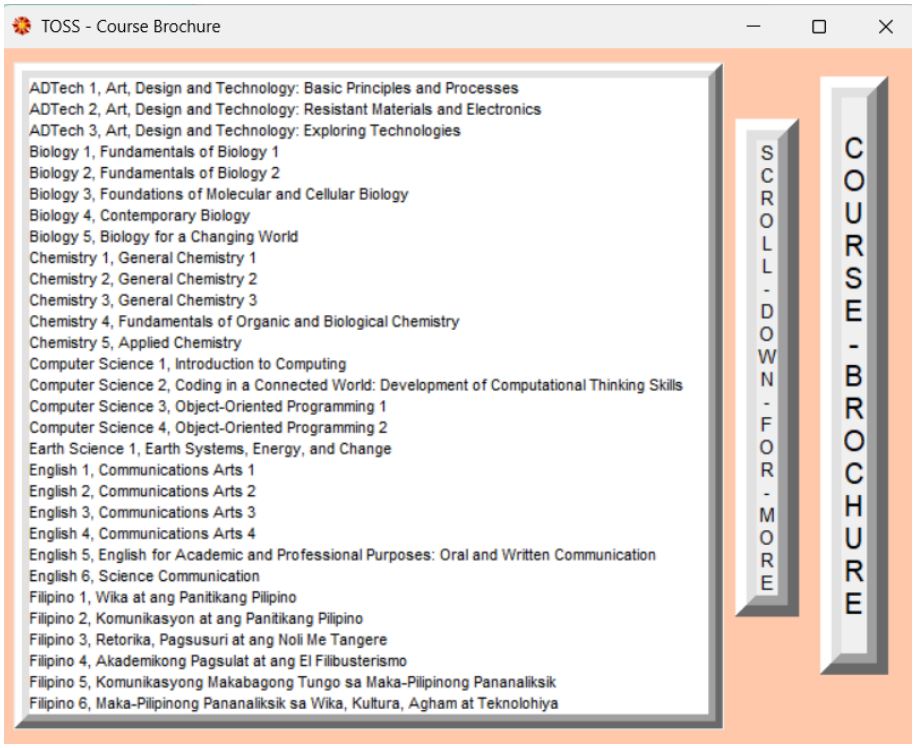
GWA Calculator (If Not Proper Inputs)



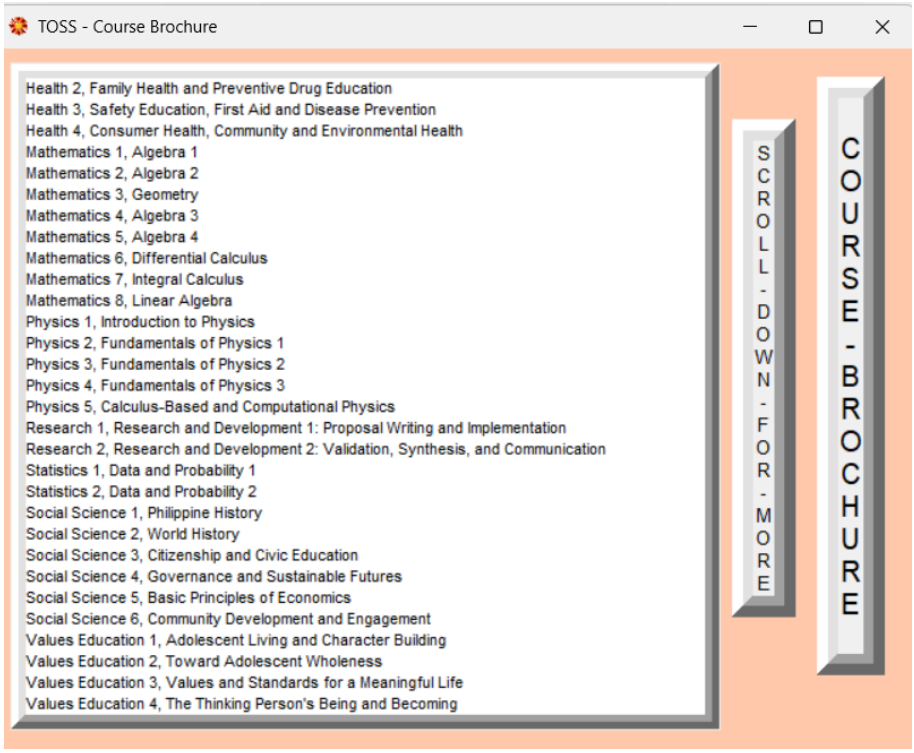
Requirements Tracker



Course Brochure (Part 1)



Course Brochure (Part 2)



Team Roles

Team	Quarter 1	Quarter 2	Quarter 3	Quarter 4
Avilla	Part of the Paper (Problem Decomposition)	Main Programmer	Main Programmer	Main Programmer and README File
Castillo	Part of the Paper (Rest of the Paper)	Flowchart of Menu Screen and Course Brochure	Programmed Course Brochure	Part of the Final Paper (Team Roles) and PPT for Defense
Manalo	Part of the Paper (Scopes and Limitations)	Part of the Paper (Problem Decomposition)	Part of the README File (Features and Program Requirements)	Part of the Final Paper (Objectives and References)
Pagilagan	Tester and Feedback Giver	Flowchart of Requirement Tracker	Part of the README FILE (How to run program, Screenshots of output, Authors/Contributors	Part of the Final Paper (Flowchart and Methodology)
Yuson	Tester and Feedback Giver	Flowchart of Requirement Tracker	Part of the README File (Project Description and Objectives)	Part of the Final Paper (Problem Statement, Project Overview and Relevance)

References / Background Research

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